CURTAIN AIR BAG MODULE [TWO-STEP DEPLOYMENT CONTROL SYSTEM]

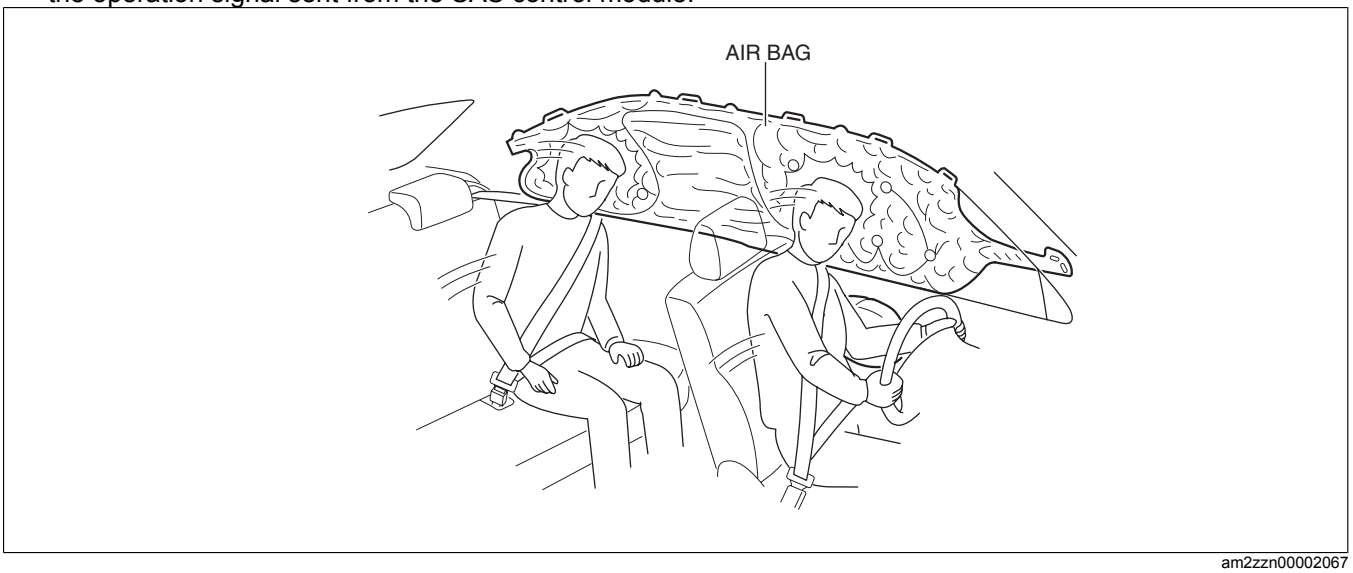
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Purpose

- When a curtain air bag module receives an impact from a lateral collision, the operation (deployment) of the air bag mediates the impact to the head of the driver or front passenger.

Function

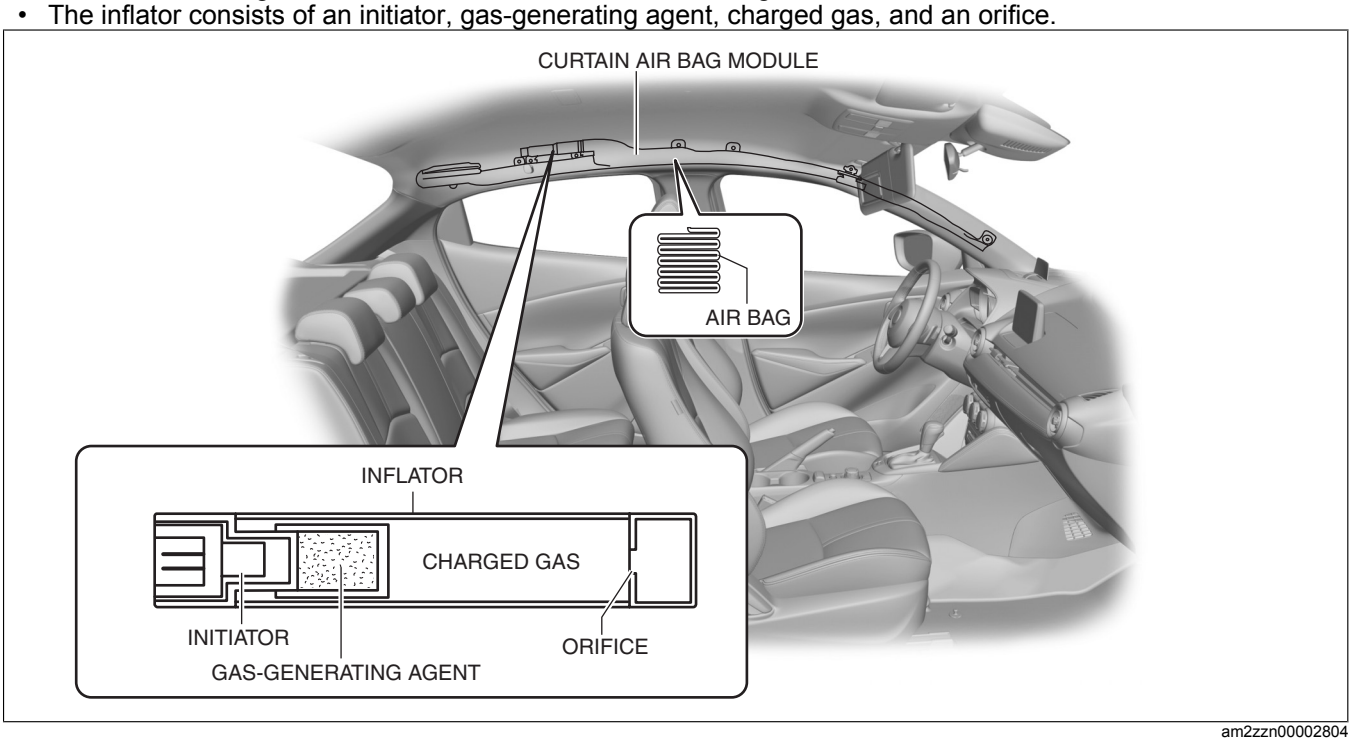
- When a curtain air bag module receives an impact from a lateral collision, the air bag is operated (deployed) by the operation signal sent from the SAS control module.



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Construction

- The curtain air bag modules are located along the roof edge between the A and C pillars.
- The curtain air bag module consists of the inflator and air bag.
- The inflator consists of an initiator, gas-generating agent, charged gas, and an orifice.

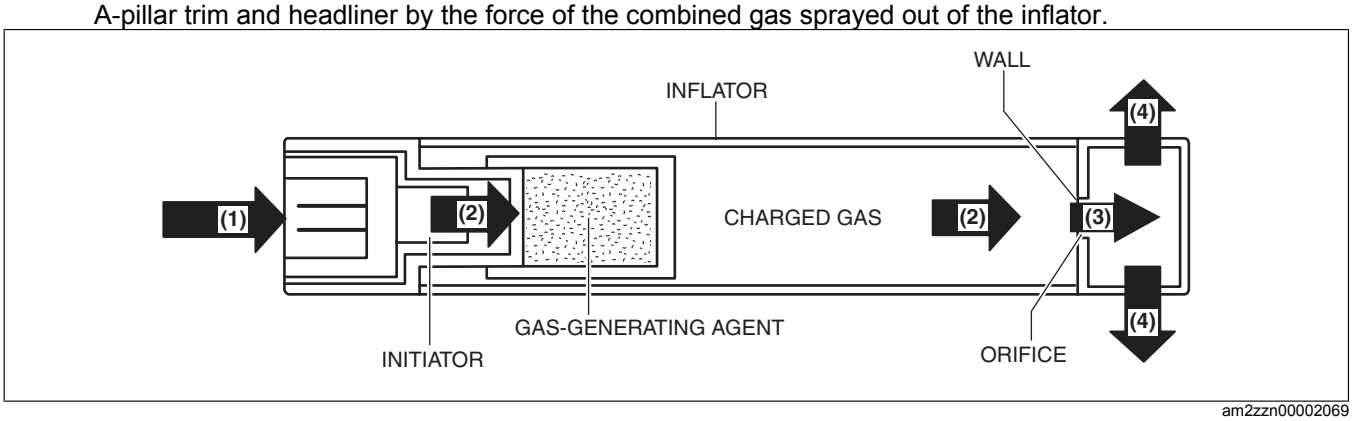


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Operation

- The curtain air bag module operates (deploys) the air bag by activating the internal inflator.
- The inflator operates (deploys) in the following order:
  1. When an operation (deployment) signal is received from the SAS control module, the initiator built into the inflator ignites.
  2. The ignition of the initiator causes the ignition of the gas-generating agent. The section of the wall from which the air bag deploys is broken by this impact wave.

3. The charged gas and the gas generated by the gas-generating agent mix and pass through the orifice.
4. The combined gas, which passed though the orifice, is injected into the air bag through the diffuser (diffusion device).
5. When the curtain air bag module operates (deploys), the air bag operates (deploys) while breaking open the A-pillar trim and headliner by the force of the combined gas sprayed out of the inflator.



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Fail-safe

- Not applicable